

RESEARCH INTERESTS

Reservoir stimulation processes, Geochemical flows, Reactive transport processes, Geological soft matter, Porous medium flows

EDUCATION

PhD in Physical Sciences

Faculty of Physics, University of Warsaw, Poland 2019 — 2025

Advisors: Prof. Piotr Szymczak and Prof. Mariusz Białecki

Thesis: Self-organization of flow in dissolving rocks

M.E. in Thermal Engineering (8 CGPA)

Thapar University, India 2015 — 2017

Advisor: Prof. Neeraj Kumar

Thesis: Nodal integral method for convection-diffusion transport in complex domain using linear and higher order quadrilateral elements

B.Tech. in Mechanical Engineering (73%)

Uttar Pradesh Technical University, India 2010 — 2014

RESEARCH EXPERIENCE

Postdoctorate fellow

Faculty of Sciences, Université libre de Bruxelles, Belgium Oct. 2025 — cont.

Advisor: Prof. Anne De Wit

Project Technical Assistant

Department of Energy Sciences, IIT Bombay Feb 2018 — Oct. 2018

Advisors: Prof. Atul Kumar and Prof. Suneet Singh

Project Research Associate

Department of Aerospace Engineering, IIT Bombay Aug. 2017 — Jan. 2018

Advisor: Prof. Manaswita Bose

AWARDS/GRANTS/FUNDING

- **FNRS-FRS mobility and congress grant** to attend Belgrade School of Complex Systems, Belgrade, Serbia 2026
- **Postdoctoral fellowship** awarded by Université Libre de Bruxelles, Belgium Oct. 2025 — Oct. 2027
- **Research fellowship for CEUS-UNISONO project** funded by National Science Centre (NCN), Ministry of Science, Poland and Slovenia Research Agency Mar. 2021 — Feb. 2025
- **Doctoral Scholarship** awarded by Interdisciplinary Doctoral School (MSD), University of Warsaw, Poland Oct. 2020 — Sep. 2024

- **Research fellowship for OPUS-11 project** funded by National Science Centre (NCN), Ministry of Science, Poland Nov. 2018 — Jan. 2021
- Travel Support by University of Paris to attend Colloquium Branching networks, France 2025
- **IDUB University research microgrant** to attend Geilo physics school, Norway 2024
- **IDUB University grant** to support completion of doctoral dissertation 2024
- **IDUB University research microgrant** to attend Geilo physics school, Norway 2023
- Early Career Scientist's Travel Support Award by European Geophysical Union (EGU), Austria 2023
- Uni. Integrated Development Programme (ZIP) travel support for a research visit Uni. of Minnesota USA. 2022
- SEG Student Chapter, Field Camp-2020, IGF-PAN 2020
- **Graduate Scholarship (GATE)**, Ministry of Education, Government of India 2015
- **University Scholarship** for Merit-Incentive Students 2010
- Bronze Medal in Programming Diploma by NCVT, Government of India 2010

JOURNAL ARTICLES

1. Deng, J., **Sharma, R. P.**, P. Szymczak, and P. K. Kang (2025). Anomalous Transport in Dissolving Porous Media: Transitions Between Fickian and Non-Fickian Regimes. *Geophysical Research Letters* 52(15).
2. Cooper, M. P., **Sharma, R. P.**, S. Magni, T. P. Blach, A. P. Radlinski, K. Drabik, A. Tengattini, and P. Szymczak (May 2023). 4D tomography reveals a complex relationship between wormhole advancement and permeability variation in dissolving rocks. *Advances in Water Resources* 175, 104407.
3. **Sharma, R. P.**, M. Bialecki, M. P. Cooper, A. P. Radliński, and P. Szymczak (June 2023). Pore merging and flow focusing: Comparative study of undissolved and karstified limestone based on microtomography. *Chemical Geology* 627, 121397.
4. **Sharma, R. P.**, J. Deng, P. K. Kang, and P. Szymczak (Oct. 2023b). Effects of Mixing at Pore Intersections on Large-Scale Dissolution Patterns and Solute Transport. *Geophysical Research Letters* 50(21).
5. **Sharma, R. P.** and N. Kumar (2018b). Nodal integral method for convection-diffusion transport using linear and higher order quadrilateral elements. *Numerical Heat Transfer, Part B: Fundamentals* 74(3), 623–645.

CONFERENCE PROCEEDINGS

1. **Sharma, R. P.** and N. Kumar (2018a). Nodal integral method for complex geometries using higher order elements. In: *Proceedings of the 24th National and 2nd International ISHMT-ASTFE Heat and Mass Transfer Conference (IHMT-2017)*. Begellhouse.

CONFERENCE POSTER/PRESENTATIONS

1. Deng, J., P. K. Kang, **Sharma, R. P.**, and P. Szymczak (n.d.). "Anomalous Transport in Dissolving Porous Media: Transitions Between Fickian and Non-Fickian Regimes". **Interpore 2025**, Albuquerque, Mexico. [Poster].
2. **Sharma, R. P.**, J. Deng, P. K. Kang, and P. Szymczak (2024). "Effects of mixing at pore intersections on large-scale dissolution patterns and solute transport". **6th Cargese summer school: FLOW and Transport in porous and fractured Media (FLOWTIME)**, Cargese, France. [Poster].

3. Deng, J., **Sharma, R. P.**, P. Szymczak, and P. K. Kang (Dec. 2023). Anomalous Transport through Dissolving Fracture Networks. In: *AGU Fall Meeting 2023, San Francisco, CA*. Vol. 2023, H12D-04, pp.H12D-04. [Poster].
4. Dzikowski, M., **Sharma, R. P.**, and P. Szymczak (2023). "High-resolution Darcy-Brinkman simulation of wormhole growth based on X-CT data". *InterPore-2023*, Edinburg, Scotland. [Poster].
5. Dzikowski, M., P. Szymczak, and **Sharma, R. P.** (2023). Empowering pre-exascale computers for Darcy-Brinkman simulation of wormhole growth based on X-CT data-can we recover experiments? In: EGU23-739. [Talk].
6. **Sharma, R. P.**, M. Bialecki, and P. Szymczak (2023). "How does chemical erosion change the pore structure of a rock?" *Geilo School 2022-The Physics of Evolving Matter: Memory, Learning and Evolution*, Geilo, Norway. [Poster].
7. **Sharma, R. P.**, M. Cooper, and P. Szymczak (2023). "Geometric measures of wormholes in dissolving rocks". *Geilo School 2023-The Physics of Evolving Matter Continued: Connectivity, Communication and Growth*, Geilo, Norway. [Poster].
8. **Sharma, R. P.**, P. K. Kang, and P. Szymczak (May 2023). Effects of mixing at pore intersections on large-scale dissolution patterns. In: *EGU General Assembly 2023, Vienna, Austria*, EGU-275, pp. EGU-275. [Talk].
9. Bialecki, M., **Sharma, R. P.**, M. P. Cooper, and P. Szymczak (May 2022). Comparative study of undissolved and karstified limestone based on microtomography. In: *EGU General Assembly 2022, Vienna, Austria*, EGU22-5229, pp. EGU22-5229. [Talk].
10. Cooper, M. P., **Sharma, R. P.**, S. Magni, P. N. H. Vu, T. Ladd, A. P. Radlinski, T. P. Blach, K. Drabik, A. Tenggattini, and P. Szymczak (Dec. 2022). Competition Between Wormholes in Dissolving Rocks Captured with 4D Tomography and Numerical models. In: *AGU Fall Meeting*. Vol. 2022, H52N-0635, pp. H52N-0635. [Poster].
11. Dzikowski, M., P. Szymczak, and **Sharma, R. P.** (Dec. 2022). Large Scale Simulations of Wormhole Growth in Dissolving Porous Media using Lattice Boltzmann Method. In: *AGU Fall Meeting*. Vol. 2022, H52N-0644, pp. H52N-0644. [Poster].
12. Lipar, M., P. Szymczak, R. Ciglič, **Sharma, R. P.**, M. Zorn, U. Stepišnik, and M. Ferik (May 2022). Challenges in characterisation and mapping of solution pipes. In: *EGU General Assembly 2022, Vienna, Austria*, EGU22-1619, pp. EGU22-1619. [Talk].
13. **Sharma, R. P.**, P. K. Kang, and P. Szymczak (2022). "The impact of intersection mixing rules on the network-scale dissolution patterns". *7th Warsaw School of Statistical Physics*, Sandomierz, Poland. [Poster].
14. **Sharma, R. P.**, M. P. Cooper, A. J. C. Ladd, and P. Szymczak (Apr. 2021). Subpixel determination of wormhole tip position in 4D tomography of dissolving limestone cores. In: *EGU General Assembly 2021, Vienna, Austria*, EGU21-14962, pp. EGU21-14962. [Poster].
15. **Sharma, R. P.** and P. Szymczak (2021). "Network modelling of wormholing process in rocks". *DREAMS21 Workshop*, University of Paris Diderot, Laboratoire Matière et Systèmes Complexe, Paris, France. [Talk].
16. Szymczak, P., M. P. Cooper, S. Magni, **Sharma, R. P.**, T. P. Blach, A. P. Radlinski, M. Dohnalik, and A. Tenggattini (Apr. 2021). Wormhole Growth in Dissolving Limestones: Insights from 4D Tomography. In: *EGU General Assembly 2021, Vienna, Austria*, EGU21-13883, pp. EGU21-13883. [Talk].
17. **Sharma, R. P.**, M. Cooper, A. J. C. Ladd, and P. Szymczak (2020). "Flowfield study in dissolved porous media". University of Wrocław, Institute of Theoretical Physics, *CFD Wrocław-7 workshop*, Wrocław, Poland. [Talk].
18. **Sharma, R. P.**, M. P. Cooper, A. J. C. Ladd, and P. Szymczak (May 2020). Modeling wormhole formation in digital rock samples: the role of segmentation and permeability-porosity relationships. In: *EGU General Assembly 2020, Vienna, Austria*. EGU General Assembly Conference, 996, pp. 996. [Poster].

19. Szymczak, P., M. P. Cooper, S. Magni, **Sharma, R. P.**, T. P. Blach, A. P. Radlinski, M. Dohnalik, and A. Tengattini (Dec. 2020). Combined Neutron and X-ray Time-Resolved Tomography of Wormhole Growth in Dissolving Limestones. In: *AGU Fall Meeting*. Vol. 2020, H081-01, pp.H081-01. [Talk].
20. Cooper, M. P., S. Magni, **Sharma, R. P.**, P. N. Vu, T. P. Blach, A. P. Radlinski, M. Dohnalik, A. Tengattini, A. Ladd, and P. Szymczak (2019). Determining the influence of pore-scale geometry on wormhole formation. In: *AGU Fall Meeting*. Vol. 2019, pp.H21M-1931. [Poster].

INVITED TALKS/SEMINARS

1. **Sharma, R. P.**, M. P. Cooper, and P. Szymczak (2025b). "Influence of rock structure on the morphology of wormhole network". **Colloquium Branching networks**: from experiment to theory, University of Paris Cite, Paris, France.
2. **Sharma, R. P.** (2023). "Physics of micro-scale cave formation in dissolving rocks". Interdisciplinary Doctoral School seminar (**MSD 2023**), Checiny, Poland.
3. **Sharma, R. P.** and P. Szymczak (2023). "Effects of mixing at pore intersections on large-scale dissolution patterns". Department of Earth and Environmental Sciences, **University of Minnesota**, USA.
4. **Sharma, R. P.**, M. Cooper, and P. Szymczak (2021a). "Fractal shapes in dissolving rocks". Symposium of young scientist (**SMN2021**), University of Warsaw, Poland.
5. **Sharma, R. P.**, M. Cooper, and P. Szymczak (2021b). "Study of dissolution channels and their geometrical properties using XCMT images". **Looping Network Meeting**, University of Paris Diderot, Paris, France.
6. **Sharma, R. P.**, M. Cooper, and P. Szymczak (2021c). "Study of geometrical measures of wormholes using 4D-tomography". **Complex system seminar**, Faculty of Physics, University of Warsaw, Poland. Nov-2021.
7. **Sharma, R. P.**, M. Cooper, and P. Szymczak (2021d). "Study of geometrical measures of wormholes using 4D-tomography". **Soft matter and statistical Physics seminar**, Faculty of Physics, University of Warsaw, Poland. Nov-2021.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Fortran

Computing: OpenFOAM, Matlab, Mathematica

TEACHING

Graduate Teaching Assistant

Hydrodynamics and Elasticity

Fall, 2023

Geophysical Laboratory-II

Spring, 2022

Student Team Project Supervisor

Study of the hydrodynamic properties of a mixture of ground coffee and water

Spring, 2022

Introduction to OpenFOAM

Spring, 2024

REFERENCES

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